• Whereas scientists say they seek knowledge for its own sake, value-free and without regard to their utility, designers value knowledge that improves the world, at least in the dimension they attend to These contrasts are telling. Regrettably, the extraordinary celebration of enlightenment science and its scientists has rendered professions that make things happen—artists, designers, engineers, managers, and agriculturalists—into practitioners of a craft, mere consumers of “pure” science, physics, for example. Herbert Simon (1969) sought to overcome the disparity of pure versus applied science by proposing a science of the artificial. Unfortunately, his proposal did not go far enough and did not manage to change this inequality. He defines design as an effort to improve a system—technological or social—but limited his concerns to rational, not human- centered choices among alternatives. I would cast the net for design more broadly and equate design, at least to start, with any effort to shape if not invent the affordances for new practices of living, forms of organization, and even languages to arise. Abstractions, by definition, are removed from everyday practices of living and can therefore easily mislead people into undesirable ventures and cognitive traps (Stolzenberg, 1984). The cybernetician W. Ross Ashby (1956) adopted a less philosophically committed definition of cybernetics. While acknowledging the circularities that are inherent in numerous phenomena—from the realization of purpose to the emergence of self—he defined cybernetics as the study of all possible systems, which is informed by what cannot be built or evolve in nature. He thus put designers and observers on an equal footing and right into the domain that cybernetics is to explore. His definition also applied the very theory of information it advanced—reducing uncertainties by experiencing constraints on possibilities—to itself, here to systems that are imaginable but reveal themselves as unrealizable. Much of Ashby’s work can be characterized as exploring the epistemological difference between synthesis and observation, between what can be designed and what can be learned from using a design without knowing its internal structure, intentions or origin. He took the notion of a “black box” that hides its makeup from its observer or experimenter as a metaphor for exploring brains, self-organizing systems, and large social phenomena such as the stock market. For example, he gave his students the task of figuring out the behavior of a non-trivial machine he had devised. As a machine, it was determinate, but its internal structure made it not obviously predictable from the relationship between its inputs, which the experimenter could set, and its outputs, which that experimenter could observe. The concept of a black box gave brain research a practical methodology as the interior of a working brain is hardly accessible and what one knows about cognition must be deduced from what the brain does. It also became a paradigm for social research, hoping that the complexities of social phenomena can be ignored in favor of overarching regularities. Ashby certainly cared for his use of language and always acknowledged the role of designers and experimenters (observers) of the systems he explored, but he never looked into the role of language and communication in conceptualizing the systems he studied. The latter was introduced by the anthropologist Margaret Mead (1968), also a participant in the Macy Conferences, who suggested that cyberneticians apply cybernetic principles to themselves, a suggestion that Heinz von Foerster (1974) coined “second-order cybernetics” and defined as the practice of including the observer in the observed. I prefer second-order cybernetics not to be limited to observers, spectators or theorists. As already mentioned, Ashby derived many cybernetic insights by exploring to the extent possible the dialectic between what